

Folding from predicted contacts instead of MSAs

Helico + Marifold · 38 homology-filtered FoldBench monomers

every target < 40% identity to Marifold's training data, and outside Helico's training window

github.com/Open-Athena/helico · PR #13, issue #11

The idea

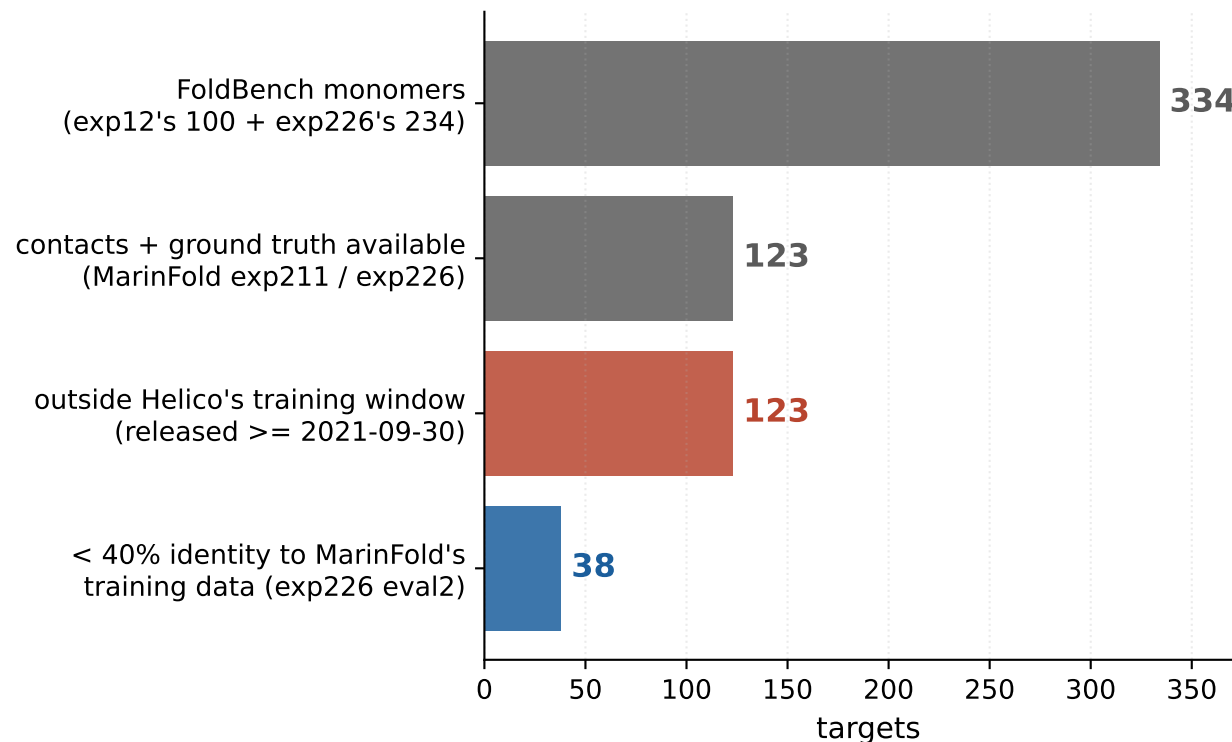
AF3-family models lean on MSAs. Strip the alignment and accuracy collapses.

- MarFold predicts residue-residue side-chain contacts directly.
- Feed those contacts to the folding model in place of the alignment, and the MSA search leaves the critical path.
- Helico takes a three-state contact matrix — contact / no-contact / unknown — added into the pair representation.
- Training samples the conditioning level per example, so one model spans no contacts through fully specified.

*Helico arms use no alignment and no conservation profile, at training or inference.
The Protenix +MSA baselines do use MSAs — that is the comparison.*

What we benchmark on, and why it is this small

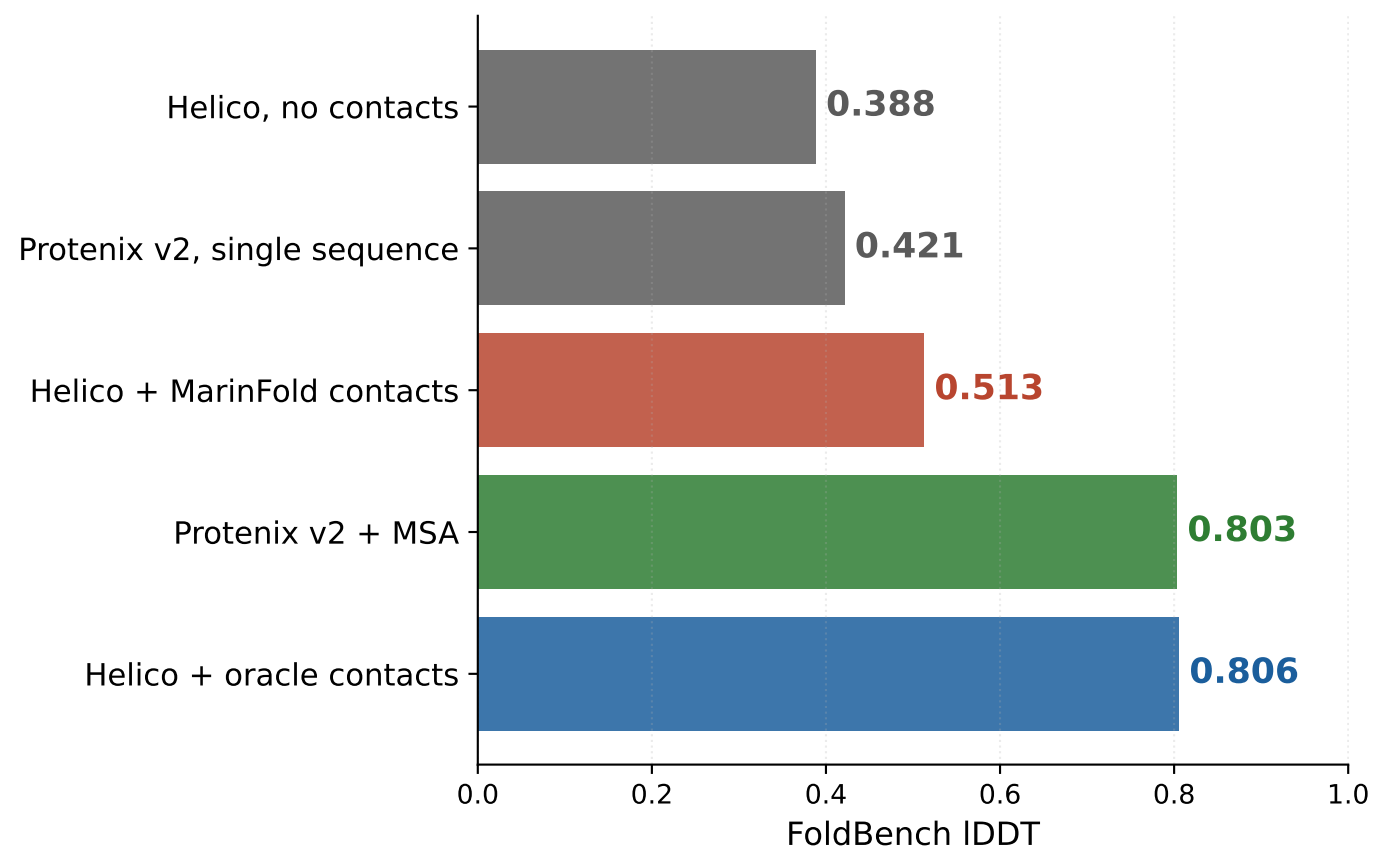
two independent filters, neither of which is optional



Only 15 of the original FoldBench 100 clear the homology filter — 85% of that set has a $\geq$ 40% homolog in MarinFold's training data. Earlier versions of this deck reported +0.229 IDDT on the unfiltered 100; that number is not shown anywhere here. CAMEO hard and CASP FM are benchd too but kept out of the headline: their depositions fall inside Protenix v2's training window, so its baselines there are optimistic.

Where things land

38 homology-filtered FoldBench monomers · Helico arms use no MSA



**Contacts beat the best
single-sequence model
 $+0.091 \pm 0.028$**

**Oracle contacts match
Protenix v2 + MSA
 -0.002 ± 0.028**

...so the ceiling is intact.
The shortfall is contact
quality, not the approach.

Predicted contacts beat single-sequence folding

against Protenix v2 — the stronger baseline — in single-sequence mode

- vs Protenix v2 single sequence: $+0.091 \pm 0.028$ IDDT ($t = 3.3$, better on 26/38)
- vs the same Helico weights with contacts withheld: $+0.124 \pm 0.029$ (better on 27/38)
- Protenix v2 is genuinely the harder baseline, and this is like-for-like: both see one sequence and no alignment.

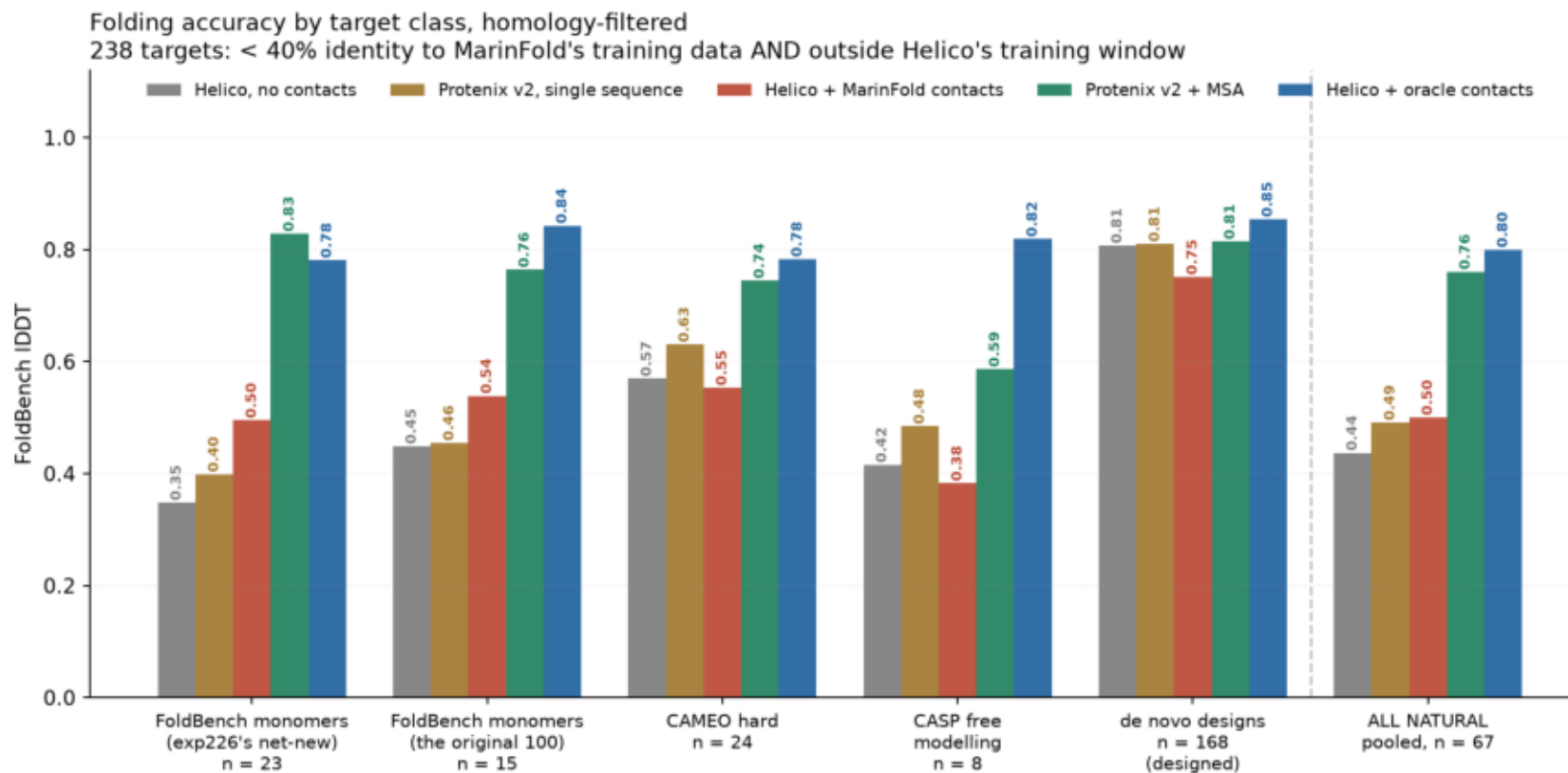
The control that makes this credible

Our own contacts-off arm scores 0.388 — BELOW Protenix v2's 0.421 (-0.033 ± 0.014).

The fine-tuned model has no intrinsic edge with no information. All of the gain comes from the contacts.

Where the gain exists, and where it does not

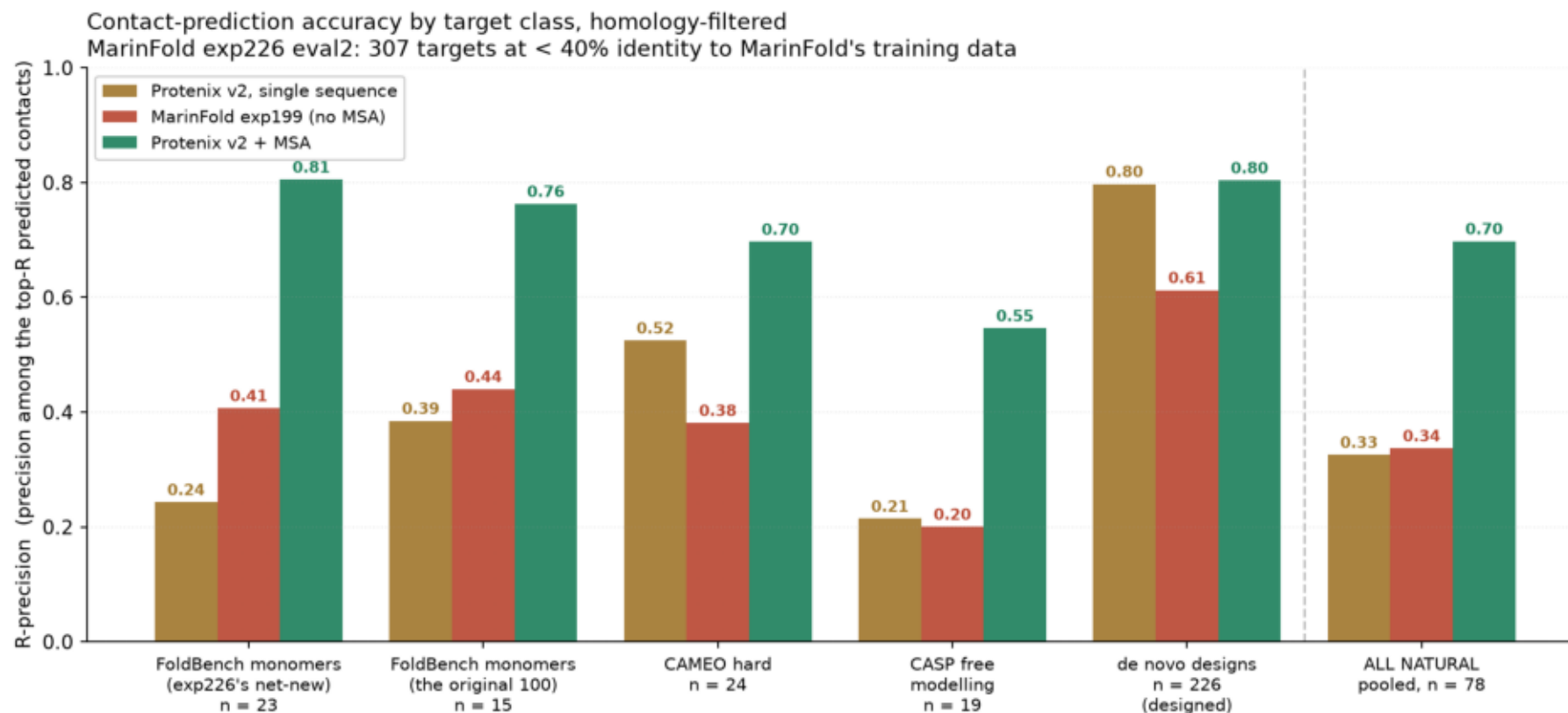
same arms, every benched class, homology-filtered throughout



The gain needs BOTH a weak single-sequence baseline AND accurate contacts. De novo designs have no headroom — Helico scores 0.81 with no contacts at all. CAMEO hard and CASP FM have headroom, but MarinFold's contacts there are too poor to claim it (R-precision 0.38 and 0.20, against 0.41–0.44 on FoldBench). Only the FoldBench slices have both — and they are also the only slices whose depositions postdate Protenix v2's training cutoff, so its bars on CAMEO hard and CASP FM read high.

Is MarinFold actually supplying better contacts?

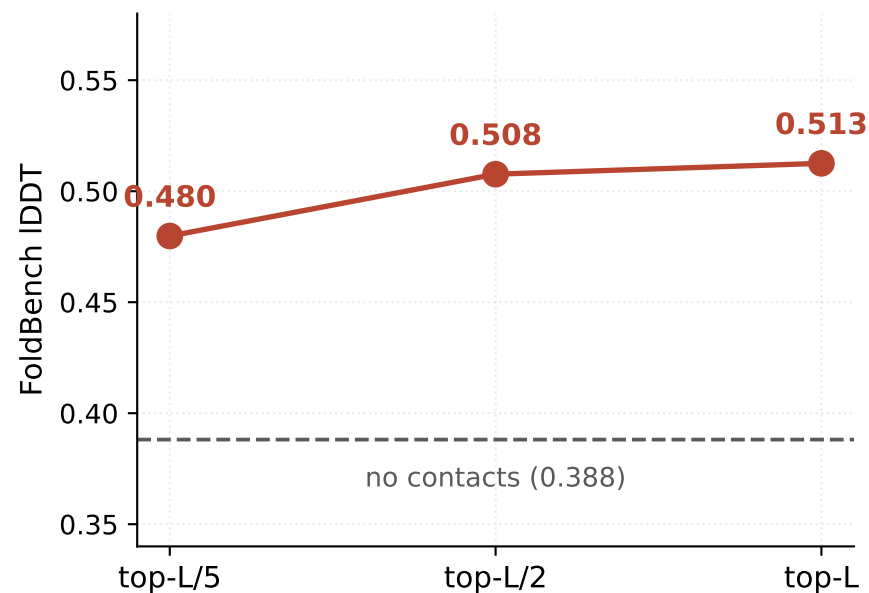
R-precision on exp226's eval2 — the same homology filter



MarinFold's own aggregate reads as a tie with Protenix v2 single sequence because 74% of eval2 is designed protein, where Protenix is much the better contact predictor. On the FoldBench slices MarinFold leads by +0.164 and +0.055 — exactly the slices where the folding gain appears. The folding model tracks contact quality faithfully.

How many contacts should MarinFold emit?

same targets, three truncation budgets



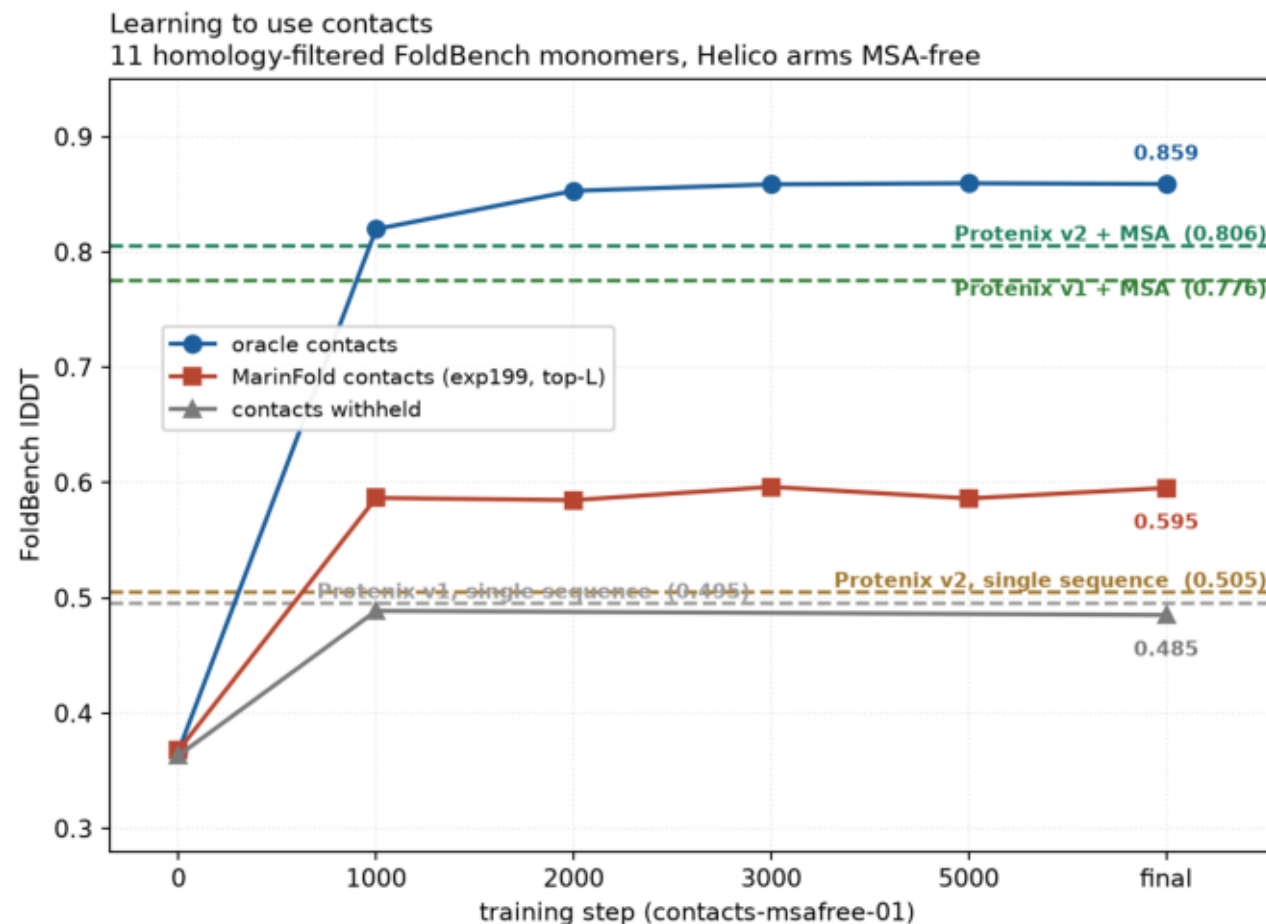
But this does not generalise off FoldBench

Pooled over all 67 filtered natural targets the trend flattens and reverses: top-L/5 0.505 vs top-L 0.500.

On designed proteins every extra contact costs accuracy monotonically. At ~0.4 precision the highest-voted fifth carries most of the true contacts, and the tail adds false positives faster than true ones.

How the model learned to use contacts

checkpoint sweep of contacts-msafree-01, benched on MarinFold contacts



Step 0 is the warm start — Protenix v1 weights, contact projection still zero-initialised, so conditioning is an exact no-op. All three arms coincide there within 0.004. Almost all of the learning happens in the first 1000 steps.

The remaining gap to MSAs is contact quality

MarinFold top-L 0.513 -> Protenix v2 + MSA 0.803

- Oracle contacts reach 0.806 against Protenix v2 + MSA's 0.803. That difference, -0.002 ± 0.028 , is not distinguishable from zero.
- A perfect contact map is worth as much as an alignment — on exactly the targets where alignments are hardest to build. Nothing about the approach caps out below MSAs.
- The entire shortfall is the quality of the contacts we can predict today: 0.513 with real ones against 0.806 with perfect ones.

What to do about it

- Improve contact accuracy on natural proteins. That is the whole gap, and R-precision on the filtered natural set is ~ 0.34 against a ceiling of 1.0.
- Retrain Helico with false positives sampled from near-miss pairs rather than uniformly — the model has never seen the error distribution it faces.
- Revisit the emitted budget. top-L is best on FoldBench but not beyond it; the right truncation depends on precision, and precision varies by class.

Caveats

$n = 38$. Homology filtering is what costs the sample size, and it is not optional — unfiltered, the same comparison reads +0.229 instead of +0.091. Warm-started from Protenix v1. Monomers only. Marifold indexes into the published prompt and Helico into resolved residues; every index map here is verified per target.